

Statistical and Machine Learning Methods for Spatial Regression

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Roadmap

- Part I: Introduction to Regression and Spatial Linear Models
- Part II: Machine Learning (ML) models in Spatial Regression
- Part III: Software Demonstration

What is Spatial Data?

Any data with some geographical information

- Geostatistics: Data represents a sample from a continuous spatial domain
- Each observation (data unit) is associated with a geographical location (latitude-longitude)
- Common sources of spatial data: climatology, forestry, ecology, environmental health, disease epidemiology, real estate marketing
- Other examples where spatial need not refer to space on earth:
 - Neuroimaging (data for each voxel in the brain)
 - Genetics (position along a chromosome)
 - Spatial transcriptomics (gene expression on slides)

What is spatial data?

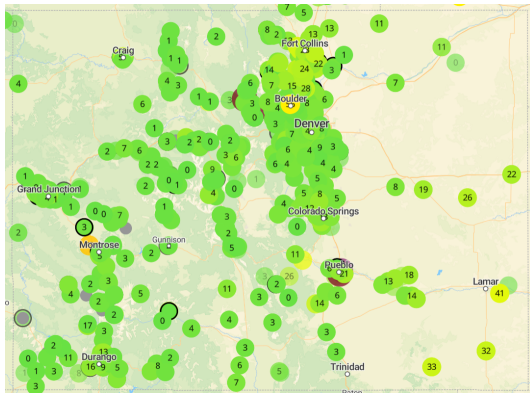


Figure: PM_{2.5} ($\mu\text{g}/\text{m}^3$) in Colorado on Sept 22, 2026 from PurpleAir.com

A familiar starting point

Point referenced data:

- Response $Y(s_i)$ and covariates $\mathbf{X}(s_i)$ observed at n spatial locations $s_1, \dots, s_n$
- Multiple covariates, moderately large n

Objectives:

- Understand the relationship between Y and X
- Predict Y at unobserved locations
- Understand spatial patterns in Y

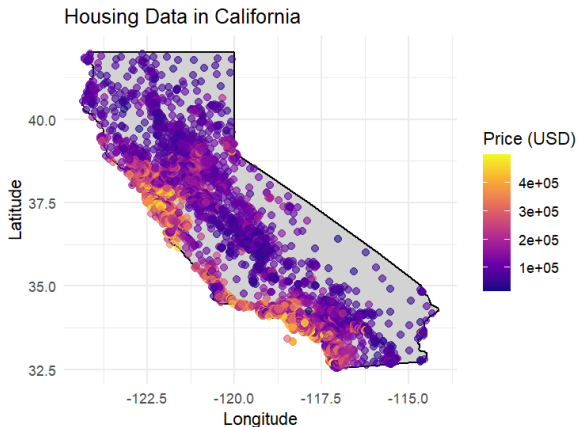
Ordinary least squares

The OLS model treats observations as independent:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

- Fit by OLS: $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
- Report coefficients, R^2 , fitted-vs-actual diagnostic plots

Example: House prices in California



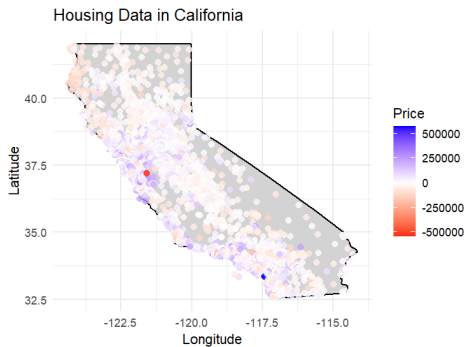
Covariates:

- Median income
- Median house age
- Total rooms
- Total bedrooms
- Population
- Number of households
- Ocean proximity

Data available on [Kaggle.com](https://www.kaggle.com/datasets/californiahousing)

Look at the residuals in space

- Map the residuals $\hat{\epsilon}(s_i)$ at their locations
- They are **not** salt-and-pepper: nearby residuals are similar
- The $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ assumption is visibly violated



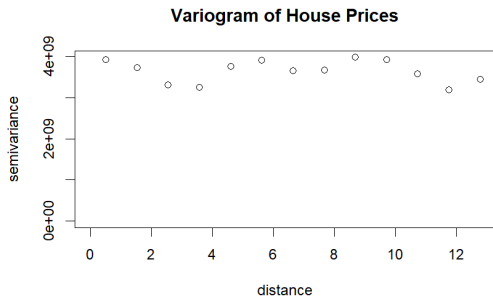
The structure OLS discards is exactly the signal we want to use.

Diagnosing the dependence

Empirical semi-variogram:

$$\gamma(h) = \text{Average of } (Y(s_1) - Y(s_2))^2 \text{ for all pairs } s_1, s_2 \text{ such that } s_1 - s_2 \approx h$$

- For spatial data, $\gamma(h)$ is expected to roughly increase with the distance h



Why it matters?

Ignoring spatial correlation has two costs:

- **Inference:** $\hat{\beta}$ stays unbiased, but standard errors are wrong (usually too small) $\Rightarrow$ inflated significance, invalid CIs
- **Prediction:** the spatially structured signal in the residuals is thrown away $\Rightarrow$ suboptimal predictions

Reframe

The correlation is not a nuisance to ignore; **it is information**. The rest of the talk is about *using* it.

Spatial linear mixed model

Add a spatial random effect to the mean:

$$Y(s) = \mathbf{X}(s)^\top \boldsymbol{\beta} + \underbrace{w(s)}_{\text{spatial}} + \underbrace{\epsilon(s)}_{\text{noise}}, \quad w(\cdot) \sim \mathcal{GP}(0, C(\cdot, \cdot; \boldsymbol{\theta})).$$

Marginally,

$$\mathbf{Y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}), \quad \boldsymbol{\Sigma} = \mathbf{C}(\boldsymbol{\theta}) + \tau^2 \mathbf{I}.$$

- C is a covariance function (e.g. Matérn); $\boldsymbol{\theta}$ = covariance parameters (variance, range, smoothness)
- Prediction at a new site: **Kriging** — a covariance-weighted combination of observations

Generalized least squares

Estimating β under correlated errors uses the covariance:

$$\hat{\beta}_{\text{GLS}} = (\mathbf{X}^\top \Sigma^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \Sigma^{-1} \mathbf{y}.$$

- Σ^{-1} “whitens” the correlated errors, so information is counted correctly
- Gives **valid inference** *and* **better prediction**
- This is the fix for the problems on the OLS residual slide

Cressie (1993); Banerjee, Carlin & Gelfand (2014).

There is no free lunch!

GLS and the likelihood need Σ^{-1} and $\det(\Sigma)$:

$$\ell(\beta, \theta) = -\frac{1}{2} \log |\Sigma| - \frac{1}{2} (\mathbf{y} - \mathbf{X}\beta)^\top \Sigma^{-1} (\mathbf{y} - \mathbf{X}\beta).$$

- Σ is $n \times n$ and dense $\Rightarrow \mathcal{O}(n^3)$ time, $\mathcal{O}(n^2)$ memory
- Fine at $n = 100$; doable at $n = 1,000$; hopeless at $n = 10^5$ – 10^6

This computational bottleneck is what motivates modern research in spatial statistics.

Scalable Gaussian processes (a pointer)

A large toolbox exists to break the $\mathcal{O}(n^3)$ computational bottleneck:

- **Low-rank / predictive process** — Banerjee et al. (2008)
- **Covariance tapering** — Furrer, Genton & Nychka (2006)
- **Nearest-neighbor / Vecchia (GpGp/NNGP)** — Vecchia (1988); Datta et al. (2016); Guinness (2019)
- **SPDE-INLA** — Lindgren, Rue & Lindström (2011)
- **Multi-resolution / LatticeKrig** — Nychka et al. (2015)

Benchmarked head-to-head in Heaton et al. (2019), *JABES* — a good one-stop reference.

One intuition: nearest-neighbor GP

Replace the full joint density by a product of small conditionals:

$$p(\mathbf{w}) = \prod_{i=1}^n p(w(s_i) \mid \mathbf{w}_{N(i)}), \quad |N(i)| = m \ll n.$$

- Each location conditions on only $m \approx 10\text{--}15$ near neighbors
- Yields a **sparse** precision matrix $\Rightarrow$ near-linear $\mathcal{O}(nm^3)$

Dropping linearity

Real covariate effects are nonlinear and interacting. Generalize the mean:

$$Y(s) = \underbrace{f(\mathbf{X}(s))}_{\text{flexible nonlinear}} + \underbrace{w(s)}_{\text{spatial GP}} + \epsilon(s).$$

- f : **Basis functions** (Diggle and Hutchinson, 1989); **GAMs** (generalized additive models; Nandy et al., 2017, *JRSSB*)
- f is a **flexible ML function** (random forest, neural net, boosting, ...)
- $w(s)$ still carries the spatial correlation

The ML model plays the role of the **mean function** in a spatial regression.

But ML is built for independent data

Off-the-shelf ML estimates f assuming the observations are i.i.d. Two consequences:

- Same invalid-inference problem as OLS
- ML can **absorb the spatial signal into** f which hurts spatial interpolation and interpretation

Two response strategies

1. **Decorrelate, then learn** — preprocess the data, use any ML model
2. **Build correlation into the learner** — change the loss / the algorithm

Strategy 1: decorrelate first (Heaton et al., 2025)

Whiten the data, then hand it to *any* ML/DL model:

$$\tilde{\mathbf{y}} = \mathbf{L}^{-1}\mathbf{y}, \quad \text{where } \Sigma^{-1} = \mathbf{L}^{\top}\mathbf{L} \text{ (Vecchia-based, sparse).}$$

- Transform motivated by the multivariate Gaussian + Vecchia approximation
- Fit ML on $\tilde{\mathbf{y}}$; back-transform / re-correlate predictions
- **Pro:** modular, model-agnostic, scalable **Con:** two-stage; needs a covariance estimate

Heaton, Millane & Rhodes (2025), *Journal of Data Science*.

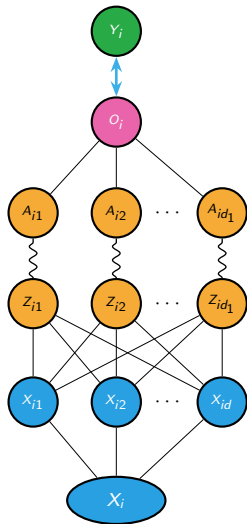
Feed-forward neural networks

$$Y_i = f(X_i) + \epsilon_i.$$

A **single-layer perceptron** models f :

$$f(X_i) = \beta^\top g_1(W_1 * X_i).$$

- Covariates X_i feed *hidden units*
- W_1, β are **weights** (unknown, estimated)
- g_1 is a **known** nonlinear *activation*
- Output layer $O_i = F(X_i)$ is fit to Y_i

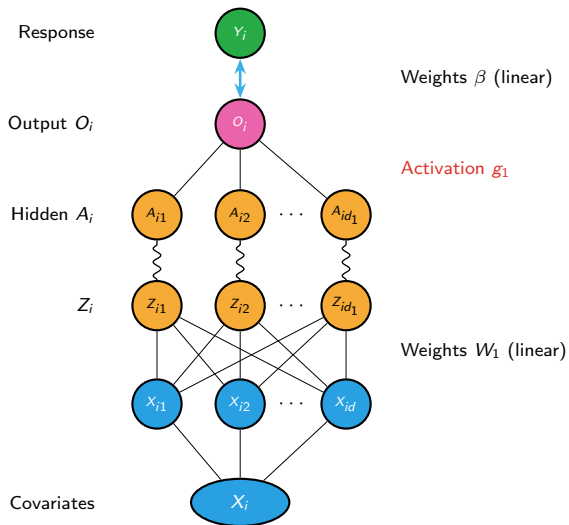


Feed-forward neural networks

$$f(X_i) = \beta^\top g_1(W_1 * X_i)$$

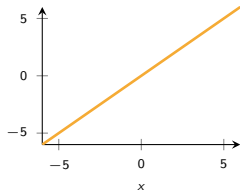
- covariates X_i
- hidden units $Z_i \rightarrow A_i = g_1(Z_i)$
- output O_i
- response Y_i

Universal approximation theorem: Any continuous function can be approximated to any degree of accuracy using a single layer perceptron with any non-polynomial activation function (Stinchcombe et al., 1989).

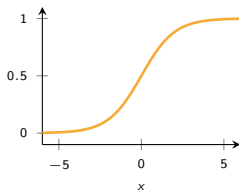


Activation functions

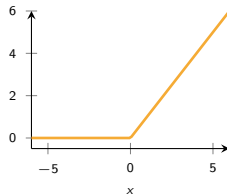
Linear



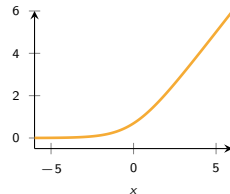
Sigmoid



ReLU



Softplus



- **Linear** output layer for a continuous response
- **Sigmoid** / **ReLU** / **softplus** in the hidden layers introduce nonlinearity

From perceptron to deep network

Multi-layer perceptron (MLP) — stack L hidden layers:

$$f(X_i) = \beta^\top g_L(W_L * g_{L-1}(W_{L-1} * \cdots g_1(W_1 * X_i) \cdots)).$$

- **Width** = units per layer; **depth** = number of layers L
- Weights $\Psi = (W_1, \dots, W_L, \beta)$ are unknown; activations g_ℓ are known

Training minimizes squared-error loss by **gradient descent**:

$$\ell(\Psi) = \sum_{i=1}^n (Y_i - O(X_i; \Psi))^2, \quad \beta^{t+1} = \beta^t - \gamma \frac{\partial \ell}{\partial \beta},$$

with **learning rate** γ .

Strategy 2: Spatial ML

(a) Add spatial features

Feed location (or basis functions of it) as extra inputs to a standard learner – **DeepKriging**

(b) Change the loss / split

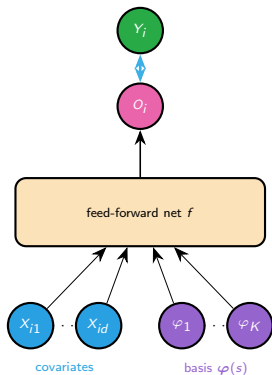
Replace the i.i.d. objective with a GLS criterion that uses Σ^{-1} – **RF-GLS, NN-GLS**

Strategy 2a: Spatial DeepKriging (Chen, Li, Reich & Sun)

Take the **same feed-forward network** — but augment the inputs with spatial basis functions $\varphi(s)$:

$$Y(s) = f\left(\underbrace{\mathbf{X}(s)}_{\text{covariates}}, \underbrace{\varphi(s)}_{\text{location}}\right) + \epsilon(s).$$

- $\varphi(s) = (\varphi_1(s), \dots, \varphi_K(s))$: multi-resolution basis functions of the coordinates
- Location enters as ordinary inputs — **standard DNN training** (the OLS loss from before)
- **Con:** spatial effect is *implicit*; can entangle with covariate effects



Chen, Li, Reich & Sun (2024), *Statistica Sinica*.

Strategy 2b: RF-GLS (Saha, Basu & Datta)

Recall trees / forests were one choice for the nonlinear mean f . Plain CART grows nodes by an *OLS* criterion; RF-GLS replaces it with a **GLS** criterion:

$$\underbrace{(\mathbf{y} - \hat{\mathbf{y}}(\text{split}))^\top (\mathbf{y} - \hat{\mathbf{y}}(\text{split}))}_{\text{OLS split (i.i.d.)}} \longrightarrow \underbrace{(\mathbf{y} - \hat{\mathbf{y}}(\text{split}))^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \hat{\mathbf{y}}(\text{split}))}_{\text{GLS split (spatial)}}.$$

- Folds spatial autocorrelation directly into node splitting
- Decorrelation step before resampling; prediction combined with kriging
- Software: `RandomForestsGLS` (R)

Saha, Basu & Datta (2023), *JASA*.

Strategy 2c: NN-GLS (Zhan & Datta)

Same MLP mean $f(X_i) = \beta^\top g_L(W_L * \dots g_1(W_1 * X_i))$ as before — only the *loss* changes.
Plain NN minimized the OLS loss; NN-GLS swaps in a GLS loss:

$$\underbrace{\sum_{i=1}^n (Y_i - O_i)^2 = \|\mathbf{y} - f_\psi(\mathbf{X})\|^2}_{\text{OLS loss (i.i.d.)}} \longrightarrow \underbrace{(\mathbf{y} - f_\psi(\mathbf{X}))^\top \Sigma^{-1} (\mathbf{y} - f_\psi(\mathbf{X}))}_{\text{GLS loss (spatial)}}.$$

- With NNGP's sparse Σ^{-1} , this is representable as a **graph neural network**
- Scales to large n ; gives kriging-style prediction and honest uncertainty
- Software: geospaNN (Python)

Zhan & Datta (2024), *JASA*.

Is AI actually ‘Super’ Intelligent? **Short answer, NO.**

Problem: predict soil moisture (SM) in the US corn belt from Pacific sea-surface-temperature anomalies (SSTA), lagged 3 months. *Skill score* = % improvement in MSPE over a trivial **persistence** forecast (positive = better than persistence).

	All months			May months		
	MSPE	Test R^2	Skill score	MSPE	Test R^2	Skill score
Persistence	3979	0.65	—	2562	0.83	—
Climatology	5116	0.54	−29%	4945	0.63	−93%
Spatial FLM	2252	0.79	43%	1880	0.86	27%
XGBoost	2642	0.74	34%	2519	0.82	2%
ANN (Raw)	5695	0.53	−43%	4995	0.66	−94%
ANN(PCA ₆₀)	5779	0.53	−45%	4690	0.69	−83%
ANN (Wide Raw)	6535	0.47	−64%	5856	0.34	−128%
ANN (Wide PCA ₆₀)	6299	0.45	−58%	5163	0.32	−101%
CNN (SSTA)	4862	0.55	−22%	4323	0.68	−69%
CNN (SSTA, SM)	3824	0.63	4%	3352	0.74	−31%

Wikle, Datta, Hari, Boone, **Sahoo**, Kavila, Castruccio, Simmons, Burr & Chang (2023), *Environmetrics*.

Software demonstration

Jump to the R session.

Food for thought

- **Interpretability.** Model-agnostic tools (variable importance, LIME, Shapley values)
Wikle, Datta, Hari, Boone, **Sahoo**, Kavila, Castruccio, Simmons, Burr & Chang (2023). An illustration of model agnostic explainability methods applied to environmental data. *Environmetrics* 34(1):e2772.
- **What if the covariates are stochastic?** We treated $X(s)$ as fixed and known. If $X(s)$ is itself a random field (or measured with error), how does that uncertainty *propagate* into the estimated mean $f(X(s))$, the spatial covariance Σ , and the prediction intervals? For $Y(s) = f(X(s)) + w(s) + \epsilon(s)$,

$$\text{Var}(Y(s)) = \underbrace{\mathbb{E}_X[\text{Var}(Y | X)]}_{\text{spatial} + \text{noise}} + \underbrace{\text{Var}_X(\mathbb{E}[Y | X])}_{\text{from stochastic } X}.$$

The second term is routinely ignored — when does it matter?

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Thank you

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